

Shujia Zhang

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SUMMARY

I want to understand quantum systems by building them.

EDUCATION

Yale University, New Haven, CT

Expected Graduation 2029

- *BS in Physics (Intensive)*
- *Relevant Coursework:* Classical Mechanics, Relativity, Quantum Information Processing, Linear Algebra, Introduction to Computer Science, Introduction to Computer Engineering, Modern Physical Measurement.
- *Awards:* 3rd place in YQuantum Hackathon - Travelers-LTImindTree-Quantinuum-Challenge.

Shanghai Starriver Bilingual School, Shanghai, China

Graduated 2025

TECHNICAL SKILLS

- **Electronics & Embedded:** Oscilloscope use, soldering, PCB reverse engineering and design (Altium Designer), PCB fabrication/prototyping (JLCPCB), embedded development (Keil/STM32), UART and serial interfaces (TTL, RS-232, RS-485), FPGA development (Vivado/Basys 3)
- **Software Engineering:** Python; codebase navigation and extension; modular transpiler design; rule-based optimization passes; implementation of algorithms from research papers; test-case design; Git/Gitee workflows; code review and debugging
- **Mechanical Prototyping:** CAD design (Fusion 360), FDM 3D printing (Bambu Lab), 3D-printed part iteration, basic mechanical assembly and fastening
- **Thermal Systems:** Peltier/TEC cooling and temperature control, air and liquid cooling, custom water-loop construction, pumps/radiators/tubing/fittings, leak testing and thermal troubleshooting
- **Scientific Computing & Data Analysis:** Experimental data collection and analysis; curve fitting and numerical analysis with NumPy, Pandas, SciPy, and Matplotlib; uncertainty/error analysis; symbolic and numerical calculations with Mathematica; Python automation for repetitive calculations and data processing

SELECTED TECHNICAL PROJECTS

Quantum Transpiler Development, *Collaborative Project*

May-June 2026

- Built a modular rule-based quantum transpiler with gate decomposition, basis translation, and configurable optimization passes. Extended an existing codebase, implemented and tested optimization routines, and incorporated software-engineer feedback to improve robustness.
- Extended a known ancilla-assisted MCX decomposition by identifying which approximate Toffoli variants preserve the required phase structure, tightening the CNOT upper bound from $16n-40$ to $16n-52$.

Diffusion Cloud Chamber, *Independent Project*

2024-Present

- Designed and built a functional diffusion cloud chamber for visualizing ionizing-particle tracks, integrating TEC cooling, closed-loop temperature control, liquid cooling, and an adjustable electric field.
- Iteratively debugged thermal performance and experimental reliability through temperature characterization, sensor evaluation, leak/sealing improvements, and mechanical redesign.

Benchmarking QAOA Against Classical Optimization Method, *Collaborative Project*

April 2026

- Implemented QAOA (Quantum Approximate Optimization Algorithm) in Python and Guppy, ran on Quantinuum's emulator Selene. Benchmarked its speed and accuracy against the classical method (PuLP CBC).
- Won 3rd place at the YQuantum Hackathon Travelers-LTImindtree-Quantinuum Challenge.

PCB/System Reverse Engineering, *Independent Project*

July-August 2026

- Reconstructed the architecture and signal flow of an undocumented commercial TEC controller, despite removed IC markings, through PCB inspection, component identification, and continuity tracing.